

# THE RANGE OF A FINITE-SCALE WEIL POSITIVITY METHOD: SUCCESS IN THE FIRST PRIME WINDOW AND ITS TERMINATION IN THE SECOND

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**ABSTRACT.** The Weil explicit formula gives an equivalent reformulation of the Riemann Hypothesis (RH) as the non-negativity of a family of local quadratic forms  $Q_W^L$ . A natural finite-scale positivity method—the *split-residual Schur criterion*—certifies  $Q_W^L > 0$  by reducing the problem to the positive-definiteness of a family of finite matrices, one per parity sector, assembled from prime-shift data. We characterise precisely the range of this method.

**First window success and margin exhaustion.** At  $L = 7/20$  (in the first prime window  $\frac{1}{2} \log 2 < L < \frac{1}{2} \log 3$ , only prime  $p = 2$  active), the method certifies  $\lambda_{\min} > 0$  with a rigorous lower bound of  $+7.81 \times 10^{-4}$  [**certify**]. A kappa-corrected pilot scan shows the window-level  $\lambda_{\min}(L) = \min(\lambda_{\min, \text{even}}^{\text{true}}, \lambda_{\min, \text{odd}}^{\text{true}})$  decreasing monotonically toward the right endpoint. Near  $L = 0.548$  the odd sector—which dominates the window minimum—is negative for  $d \leq 100$  and barely positive ( $+5.7 \times 10^{-9}$ ) only at  $d = 120$  [**pilot**]. The certify-grade interval width would exceed this signal by a factor  $\sim 5 \times 10^4$ , making the first-window right end certify-indeterminate [**pilot**].

**Second window termination.** In the second prime window  $\frac{1}{2} \log 3 < L < \log 2$ , both primes  $p = 2, 3$  are active and introduce a qualitatively new cross-prime coupling  $J(\tau_2, \tau_3)$  absent in the first window. We certify that  $J(\tau_2, \tau_3) \neq 0$  [**certify**] but that this coupling neither dominates the form nor delays its collapse—two natural conjectures, both refuted by independent adjudication. The odd sector is certifiably negative-definite even at the most favourable left end  $L = 11/20$  [**certify**], and a pilot scan records  $\lambda_{\min}^{\text{true}}(L) < 0$  throughout the window [**pilot**].

**Window-endpoint sequence and gradual exhaustion.** The sequence of right-endpoint values  $\lambda_{\min}(\frac{1}{2} \log p_k^-)$  reads: pilot knife-edge / certify-indeterminate (window 1)  $\rightarrow$  certify-negative  $\approx -0.37$  (window 2, [**certify**] anchor)  $\rightarrow \lesssim -0.7$  (window 3, [**extrapolation**]). The termination is a *gradual* process: the positive margin is continuously exhausted by the growing Weil constant  $c_L$  within window 1, then pushed unambiguously negative by prime 3’s entry at the start of window 2.

**Conclusion.** This paper characterises the complete range of the split-residual Schur method as the first prime window  $[\frac{1}{2} \log 2, \frac{1}{2} \log 3]$ . This is an honest negative result: the method does not extend to the second window, and the mechanism explains why no finite per-window extension would reach RH. No statement about RH is made. No extrapolation past  $L = \log 2$  is attempted.

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## 1. INTRODUCTION

**1.1. The Weil explicit formula and RH.** For a Schwartz test function  $f$  with support in  $(-L, L)$ , the *Weil explicit formula* expresses a global sum over non-trivial zeros  $\rho$  of  $\zeta(s)$  in terms of an archimedean integral and a sum over primes. Weil’s criterion states:

$$\text{RH} \iff Q_W^L(f) \geq 0 \quad \text{for all } f \in \mathcal{S}(\mathbb{R}) \text{ with } \text{supp } \hat{f} \subset [-L, L], \forall L > 0.$$

Defining

$$\lambda(L) := \inf_{\|f\|=1} Q_W^L(f),$$

the equivalence becomes  $\lambda(L) \geq 0$  for all  $L > 0$ .

**1.2. The prime windows.** Prime  $p$  enters the explicit formula at  $L = \frac{1}{2} \log p$ , where its shift parameter  $\tau_p = \log p/L$  crosses 2 (the single-hop regime boundary). This creates a sequence of intervals:

- Window 1:  $L \in (\frac{1}{2} \log 2, \frac{1}{2} \log 3) \approx (0.347, 0.549)$ , only  $p = 2$  active;
- Window 2:  $L \in (\frac{1}{2} \log 3, \log 2) \approx (0.549, 0.693)$ ,  $p = 2, 3$  both active;
- Window  $k$ : bounded by primes  $p_k, p_{k+1}$ .

Inside each window the set of single-hop primes is constant, so the combinatorial structure of the positivity problem is fixed.

**1.3. The split-residual Schur method.** We work in a Legendre basis  $\{P_n\}_{n \geq 0}$  on  $L^2(-L, L)$ , truncated at degree  $N$  per parity sector. In each sector the method assembles a finite Schur matrix

$$C = b_L \cdot F - R_\eta,$$

where  $b_L = H_d - c_L - \kappa_L$  is the Archimedean reserve ( $H_d$  the  $d$ -th harmonic number,  $c_L = \log(2\pi L) + \gamma_E$  the Weil constant,  $\kappa_L$  a bounded correction),  $F = T + M^{(0)} + M^{(2)} - c_L G$  encodes the Fourier–prime–cross structure, and  $R_\eta$  is a residual from the split. The second-moment object  $M^{(2)}$  must include all four terms  $S_{VV} + S_{VK} + S_{KV} + S_{KK}$ ; omitting any term inflates the apparent positivity (a documented defect class).

**Positivity criterion.**  $C$  positive-definite (equivalently, all  $\text{LDL}^\top$  pivots  $> 0$ ) is a *sufficient* condition for  $\lambda(L) > 0$  in the corresponding sector. A negative minimum pivot proves only that *this certificate* fails—not that  $\lambda(L) \leq 0$ .

1.4. **Evidence grades.** Every numerical statement in this paper carries one of three grades:

- **[certify]** — outward-rounded interval arithmetic (Arb balls); the sign verdict is made only when both interval endpoints lie on the same side of zero.
- **[pilot]** — float-center computation, no interval enclosure; directional and shape evidence only, never a final verdict.
- **[extrapolation]** — extrapolation beyond computed windows; indicative only, not a certified bound.

No finite sample of  $L$ -values is promoted to a whole-window assertion.

### 1.5. Main result (informal).

**Theorem 1.1** (Range characterisation, informal). *The split-residual Schur method certifies  $\lambda(L) > 0$  at  $L = 7/20$  in the first prime window [certify]. A pilot scan confirms positive sign across the sampled interior of the first window, with margin collapsing to  $O(10^{-8}-10^{-9})$  near the right endpoint [pilot]. In the second prime window the method fails: the odd sector is certifiably not positive-definite at  $L = 11/20$  [certify]. The window-endpoint infimum sequence decreases monotonically through zero. The method’s range is contained in the first prime window. No statement about RH follows.*

**Organisation.** Section 2 covers the first window. Section 3 covers the second window and the cross-prime structure. Section 4 describes the endpoint sequence. Section 5 states the range termination and its methodological implications. Appendix A is the complete data-provenance table.

## 2. THE FIRST PRIME WINDOW

### 2.1. Certification at $L = 7/20$ .

**Proposition 2.1** (FP-0.35 [certify]). *At  $L = 7/20$ , using the full four-term  $S^{(0)} = S_{VV} + S_{VK} + S_{KV} + S_{KK}$ , real Weil constant  $c_L = 1355726/993009 \approx 1.36527$ , and parameters  $N_{\text{even}} = 8$ ,  $d_{\text{even}} = 16$ ,  $N_{\text{odd}} = 6$ ,  $d_{\text{odd}} = 13$ :*

$$\lambda_{\text{lb}}(7/20)_{\text{even}} \geq +7.81 \times 10^{-4}, \quad \lambda_{\text{lb}}(7/20)_{\text{odd}} \geq +4.92 \times 10^{-2}.$$

*Both sectors are certified positive-definite.*

**Remark 2.2.** The certified lower bound  $\lambda_{\text{lb}}$  may be substantially smaller than the true minimum eigenvalue  $\lambda_{\text{min}}^{\text{true}}$ . At  $L = 7/20$  we have  $\lambda_{\text{min,even}}^{\text{true}} \approx +9.5 \times 10^{-4}$  and  $\lambda_{\text{min,odd}}^{\text{true}} \approx +1.90 \times 10^{-2}$  [pilot], so the certificate is conservative by roughly  $1.2\times$  (even) and  $2.6\times$  (odd). This distinction matters at the right end of the window; see ?? below.

**Remark 2.3** (Historical defect). An earlier certificate used  $S^{(0)} = S_{KK}$  only (three terms omitted), which inflated the reported eigenvalue by approximately  $16\times$  and yielded a false  $\lambda_{\text{lb}} \approx +0.015$ . The corrected certificate uses all four terms and is the one cited here. Source: `pilots/cert_fp035_clean.json` (weil-first-prime).

2.2. **Shape of  $\lambda_{\text{min}}^{\text{true}}(L)$  across the first window. Note on a corrected computation.** An earlier scan of this window hard-coded  $\kappa = 1.25528305$  (the value at  $L = 7/20$ ) for all  $L$ , instead of evaluating  $\kappa(L)$  per-point. The scan has been rerun with the correct per- $L$   $\kappa$ . The data in this section are the corrected values. The FP-0.35 certificate at  $L = 7/20$  is *not* affected (that point used the correct  $\kappa$  throughout).

We sampled  $\lambda_{\text{min}}^{\text{true}}(L)$  (float-center minimum eigenvalue) for the even sector at  $d = 60$  and for the odd sector at  $d \in \{60, 80, 100, 120\}$ , at selected points across the window.  $\lambda_{\text{min}}(L) := \min(\lambda_{\text{min,even}}^{\text{true}}, \lambda_{\text{min,odd}}^{\text{true}})$  is the window-level quantity.

TABLE 1. First-window shape: even sector at  $d = 60$  and odd sector  $d$ -scan at  $L = 0.548$  ([**pilot**], kappa-corrected).  $\lambda_{\min}$  is dominated by the *odd* sector near the right end.

| $L$   | Even $\lambda_{\min}^{\text{true}} (d = 60)$ | Odd $\lambda_{\min}^{\text{true}}$ | Window $\lambda_{\min}$               |
|-------|----------------------------------------------|------------------------------------|---------------------------------------|
| 0.350 | $+2.6 \times 10^{-3}$                        | far positive                       | even-dominated, positive              |
| 0.430 | $+5.9 \times 10^{-5}$                        | positive                           | even-dominated, positive              |
| 0.490 | $+1.2 \times 10^{-6}$                        | positive                           | even-dominated, positive              |
| 0.510 | $+3.7 \times 10^{-7}$                        | positive                           | even-dominated, positive              |
| 0.548 | $+5.2 \times 10^{-9}$                        | see below                          | <b>odd-dominated</b> (see Remark 2.4) |

Odd sector at  $L = 0.548$  (right end),  $d$ -scan [**pilot**]:

|           |                       |                              |
|-----------|-----------------------|------------------------------|
| $d = 60$  | $-1.5 \times 10^{-5}$ | negative                     |
| $d = 80$  | $-6.1 \times 10^{-6}$ | negative                     |
| $d = 100$ | $-2.4 \times 10^{-6}$ | negative                     |
| $d = 120$ | $+5.7 \times 10^{-9}$ | knife-edge positive (barely) |

**Remark 2.4** (Right-end knife-edge and certify-indeterminate status). Near  $L = \frac{1}{2} \log 3 \approx 0.549$ , the window-level  $\lambda_{\min}$  is *dominated by the odd sector*, which is more negative than the even at every sampled point near the right end.

**Odd sector at  $L = 0.548$ :** the value is negative at  $d \leq 100$  and barely positive ( $+5.7 \times 10^{-9}$ ) only at  $d = 120$  [**pilot**]. The convergence is extremely slow ( $d$  must triple from 40 to 120 to cross zero), and the signal of order  $10^{-9}$  is far smaller than the certify-grade interval width: the  $\tau_p$  interval arithmetic uncertainty alone is of order  $2.7 \times 10^{-4}$ , exceeding the signal by a factor  $\sim 5 \times 10^4$ .

**Even sector at  $L = 0.548$ :**  $+5.2 \times 10^{-9}$  at  $d = 60$  [**pilot**]—a knife-edge positive value, but the even sector is *not* the window minimum.

**Honest reading.** At the first-window right end:

- The even sector is knife-edge positive in the pilot [**pilot**], not certify-graded.
- The odd sector is *negative* at  $d \leq 100$  and barely positive only at  $d = 120$  [**pilot**].
- The window minimum  $\lambda_{\min} = \min(\text{even}, \text{odd})$  is at pilot-level knife-edge or negative; a certify-grade computation would be **indeterminate** (interval width  $\gg$  signal by  $\sim 5 \times 10^4$ ).

We do *not* claim a certified positive right-end value. The right end of the first window already shows margin exhaustion at pilot level.

**Shape summary ([**pilot**], kappa-corrected, sampled grid).**

- Even-sector  $\lambda_{\min}^{\text{true}}(L)$  is monotone decreasing at  $d = 60$ , from  $+2.6 \times 10^{-3}$  at  $L = 0.35$  to  $+5.2 \times 10^{-9}$  at  $L = 0.548$ .
- Odd-sector  $\lambda_{\min}^{\text{true}}(L)$  decreases similarly; at the right end it converges to the pilot level only at  $d = 120$  and is negative for  $d \leq 100$ .
- The window minimum is **odd-dominated** at the right end.
- At  $L = 7/20$  (interior), the certified odd bound is  $+4.92 \times 10^{-2}$  [**certify**], far from zero.

We do not assert a certified whole-window monotonicity result; the above is a sampled-grid description.

### 3. THE SECOND PRIME WINDOW

**3.1. New structure: the cross-prime coupling.** In the second window  $\tau_2 = \log 2/L$  and  $\tau_3 = \log 3/L$  are both in the single-hop regime ( $\tau_p < 2$ ). The second-moment object gains a genuinely new term absent from the first window:

$$J(\tau_2, \tau_3)_{ij} := \langle C_{\tau_3,1} P_j, C_{\tau_2,1} P_i \rangle,$$

where  $C_{\tau,1}$  is the single-hop shift operator. This coupling is algebraically distinct from the diagonal prime contributions and creates a non-trivial interaction between the two shifts.

**Proposition 3.1** (Cross-term non-vanishing [certify]). *At  $L = 3/5$ , with  $N_{\text{even}} = 8$ ,  $N_{\text{odd}} = 7$ , using outward-rounded Arb interval arithmetic:*

$$\max_{i,j} |\Delta C_{ij}| \geq \begin{cases} 0.630 & \text{even sector,} \\ 0.140 & \text{odd sector.} \end{cases}$$

Here  $\Delta C = C_{\text{full}} - C_{\text{cross-off}}$  is the cross-term contribution matrix. In particular  $J(\tau_2, \tau_3) \neq 0$ .

This is the paper’s *positive* certified result about the second window. It is bounded to non-vanishing and says nothing about sign, dominance, or positivity.

**3.2. Two refuted conjectures about the cross term.** Two natural strengthenings of Proposition 3.1 were investigated and both were refuted by independent recomputation.

**Proposition 3.2** (Dominance refuted). *The conjecture that  $J(\tau_2, \tau_3)$  dominates  $\lambda_{\min}(C)$ —i.e., that the cross-on assembly is positive-definite—is false. An outsourced report claimed Layer-1 eigenvalue  $+0.077$  under the cross-on assembly at  $L = 3/5$  even. Independent recomputation on the project’s trusted assembly gives  $\lambda_{\min, \text{even}}^{\text{true}} = -0.231$  [pilot] (sign reversed). The claimed value is arithmetically inconsistent with the independently certified min-pivot  $= -0.0197$  [certify] at the same point: a matrix cannot simultaneously have minimum eigenvalue  $> 0$  and minimum pivot  $< 0$ .*

**Proposition 3.3** (Delay refuted). *The conjecture that the cross term delays the loss of positivity—i.e., that  $\Delta V := d\lambda_{\min}(C_{\text{full}})/dL - d\lambda_{\min}(C^{(1)})/dL > 0$ —is false. At five test points in the second window, three give strongly negative  $\Delta V$  (specifically  $\Delta V = -2.74, -0.40, -2.25$  at  $L = 0.552, 0.580, 0.600$  respectively [pilot]), far outside any numerical-error margin. A single counter-example is sufficient to falsify the conjecture; three are present.*

**Remark 3.4.** Propositions 3.2 and 3.3 do *not* imply that  $J(\tau_2, \tau_3) = 0$  or is negligible. The cross term is certified nonzero (proposition 3.1) and contributes positively to the pivot at interior points (turning cross off lowers the even pivot from  $-0.036$  to  $-0.231$  at  $L = 0.60$  [pilot]). The refuted claims are about *dominance* and *delay*, not about existence.

**3.3. Per-prime decomposition at  $L = 3/5$ .** Table 2 shows the pivot contributions from each prime and the cross term at  $L = 3/5$  ([pilot]).

Prime 3 is the dominant new negative contribution (odd sector); prime 2 is nearly inert (even) or secondary (odd); the cross term is positive but insufficient to overcome the combined prime contributions.

TABLE 2. Pivot decomposition at  $L = 3/5$  [**pilot**].  $\Delta_p = \text{min-pivot}(\text{full}) - \text{min-pivot}(\text{prime-}p \text{ off})$ .

| Sector           | Full pivot | $\Delta_{\tau_2}$ | $\Delta_{\tau_3}$  | $\Delta_{\text{cross}}$ |
|------------------|------------|-------------------|--------------------|-------------------------|
| Even ( $N = 8$ ) | -0.0363    | +0.0001 (inert)   | -0.0112            | +0.1951                 |
| Odd ( $N = 7$ )  | -0.1573    | -0.0436           | -0.1130 (dominant) | +0.0999                 |

### 3.4. Certified non-positivity.

**Proposition 3.5** (Second window not positive-definite [**certify**]). *At  $L = 11/20 = 0.55$  (the left end of the second window, most favourable for positivity):*

$$\text{min-pivot}_{\text{odd}, N=7} \in [-2.815 \times 10^{-2}, -1.720 \times 10^{-2}],$$

so the odd sector is certifiably not positive-definite.

At  $L = 69/100 = 0.69$  (near the right end), the odd sector has  $\lambda_{\text{min}, \text{odd}}^{\text{true}} = -0.369$  [**certify**] (certify-anchored eigenvalue; pivot straddle confirms negative-definite).

**Remark 3.6.** The even sector at  $L = 11/20$  has min-pivot interval  $[-3.87 \times 10^{-4}, +2.43 \times 10^{-4}]$  [**certify**]—straddling zero. This is *indeterminate*: neither positive-definite nor certifiably negative. We do not claim the even sector is certifiably negative.

**3.5. Shape across the second window.** A pilot scan at  $N_{\text{even}} = 8$  ( $d = 16$ ) and  $N_{\text{odd}} = 7$  ( $d = 15$ ) over 7–9 sample points each gives:

**Odd sector.** Monotone decreasing from  $-0.118$  at  $L = 0.55$  to  $-0.369$  at  $L = 0.69$ . No flattening; infimum at the right end.  $d$ -refinement (at  $L \in \{0.66, 0.67, 0.69\}$ ,  $d \in \{12, 16, 20\}$ ) confirms the monotone descent is  $d$ -robust; the odd right-end infimum is the dominant behaviour [**pilot**].

**Even sector.** Steep descent from  $-0.093$  at  $L = 0.55$  to approximately  $-0.283$  at  $L \approx 0.65$ – $0.67$ , then a very shallow basin. The fine structure of the basin (interior minimum vs. right-end turn-up) is *not*  $d$ -robust: the apparent interior minimum shifts with  $d$  and inverts at  $d = 20$ . The shallow fine structure is a Gibbs truncation artefact and is not reported as a result. What survives: steep descent at the left end, all sampled values negative [**pilot**].

**Window  $\lambda_{\text{min}}$ .** The odd sector dominates (is more negative) at every sampled  $L$ . The window infimum over sampled points is  $-0.369$  at  $L = 0.69$  [**pilot**].

## 4. THE WINDOW-ENDPOINT SEQUENCE AND TERMINATION MECHANISM

**4.1. Right-endpoint infima.** Define  $\Lambda_k := \lambda_{\text{min}}^{\text{true}}((\frac{1}{2} \log p_{k+1})^-)$ , the infimum of the window-level  $\lambda_{\text{min}}(L) = \min(\lambda_{\text{min}, \text{even}}^{\text{true}}, \lambda_{\text{min}, \text{odd}}^{\text{true}})$  near the right endpoint of window  $k$ .

TABLE 3. Window right-endpoint infimum sequence.

| Window | Right endpoint                     | $c_L$ (right end) | $\Lambda_k$                              | Grade                     |
|--------|------------------------------------|-------------------|------------------------------------------|---------------------------|
| 1      | $\frac{1}{2} \log 3 \approx 0.549$ | $\approx 1.82$    | pilot knife-edge / certify-indeterminate | [ <b>pilot</b> ]          |
| 2      | $\log 2 \approx 0.693$             | $\approx 2.05$    | $\approx -0.37$                          | [ <b>certify</b> ] anchor |
| 3      | $\frac{1}{2} \log 5 \approx 0.805$ | $\approx 2.20$    | $\lesssim -0.7$                          | [ <b>extrapolation</b> ]  |

At window 1’s right end the margin exhaustion is already visible at pilot level: the odd sector reads  $-1.5 \times 10^{-5}$  to  $-2.4 \times 10^{-6}$  for  $d \leq 100$  and reaches a knife-edge  $+5.7 \times 10^{-9}$  only at  $d = 120$  [**pilot**]; the certify interval width ( $\sim 2.7 \times 10^{-4}$ ) would drown any

signal of this magnitude by a factor  $\sim 5 \times 10^4$ , making window 1's right end certify-indeterminate [**pilot**].  $\Lambda_2 \approx -0.37$  is a certify-anchored eigenvalue (proposition 3.5).  $\Lambda_3$  is [**extrapolation**] and has not been computed.

The sequence therefore reads: pilot knife-edge (window 1)  $\rightarrow$  certify-negative  $-0.37$  (window 2)  $\rightarrow$  extrapolated  $\lesssim -0.7$  (window 3). The *transition from window 1 to window 2 is not a sudden jump*: margin exhaustion begins already at the right end of window 1, and becomes unambiguous at the left end of window 2 (Proposition 3.5).

**4.2. Mechanism.** Three structural factors drive the sequence downward:

- (i) **Growing Weil constant.**  $c_L = \log(2\pi L) + \gamma_E$  increases from  $\approx 1.37$  at  $L = 7/20$  to  $\approx 1.82$  at the first-window right end,  $\approx 2.05$  at the second-window right end. Since  $b_L = H_d - c_L - \kappa_L$ , the Archimedean reserve  $b_L$  shrinks continuously as  $L$  increases *within* a window, driving  $\lambda_{\min}$  downward before any new prime enters.
- (ii) **Each new prime adds a negative contribution.** Prime 3's entry at the start of window 2 adds  $\Delta_{\tau_3} \approx -0.11$  to the odd pivot (Table 2). Prime 5 in window 3 would add a further negative term. This is the *discrete* jump between windows; the *continuous* drain within a window is item (i).
- (iii) **Cross terms are positive but non-dominant.** The cross-prime coupling  $J(\tau_2, \tau_3)$  contributes positively ( $+0.10$ – $+0.20$  at  $L = 0.60$ ) but does not overcome the combined negative pressure from the growing  $c_L$  and the new prime contribution (Propositions 3.2 and 3.3).

The picture that emerges is one of *gradual, continuous margin exhaustion*: the positive margin shrinks throughout the interior of window 1 (driven by item (i)), arrives at pilot-level knife-edge / certify-indeterminate at the right end of window 1, and crosses into certifiably negative territory already at the left end of window 2 (item (ii) contributing the discrete step). Window 1 is the unique window where the method succeeds because it is the only window whose *left end* opens with sufficient reserve before the continuous drain and discrete jumps accumulate past zero.

**Remark 4.1** (Window 3 is [**extrapolation**]). The claim  $\Lambda_3 \lesssim -0.7$  is an extrapolation from the trend in Table 3; it has not been computed. We record it for orientation only. No certified statement about window 3 is made in this paper.

## 5. CONCLUSIONS

### 5.1. Range of the method.

**Theorem 5.1** (Range characterisation). *The split-residual Schur positivity method has the following certified range:*

- **Succeeds** at  $L = 7/20 \in (\frac{1}{2} \log 2, \frac{1}{2} \log 3)$ : both sectors certified positive-definite [**certify**].
- **Fails** in the second window: the odd sector at  $L = 11/20$  (the most favourable point in window 2) is certifiably not positive-definite [**certify**].

*The termination is not a sudden jump at the window boundary. A pilot scan of the first window (kappa-corrected) shows the window-level  $\lambda_{\min}$  arriving at pilot-level knife-edge near  $L \approx 0.548$ , with the odd sector negative for  $d \leq 100$  and barely positive only at  $d = 120$  [**pilot**]. A certify-grade computation at the first-window right end would be indeterminate (interval width  $\gg$  signal by  $\sim 5 \times 10^4$ ). The positive margin is continuously exhausted within window 1 (growing  $c_L$ ) and pushed into certifiably negative territory at the start of window 2 (prime 3 entry). These pilot conclusions require certify-grade confirmation to be definitive.*

5.2. **What this does not imply.** We state explicitly what the above does *not* say:

- **Not a statement about RH.** The method’s range is a property of the *finite-scale construction*, not of the infinite-dimensional Weil operator. The Riemann zeta zeros (known to satisfy RH to great height numerically) are not the object under study. Failure of a sufficient criterion does not imply  $\lambda(L) \leq 0$ .
- **Not a whole-window certified result.** Only finitely many  $L$ -points have been certified. A continuous positivity/negativity result for all  $L$  in a window would require additional (e.g., Lipschitz or analyticity-based) continuation arguments.
- **Not an extrapolation past  $L = \log 2$ .** The framework changes at  $L = \log 2$  (the fourth prime  $p = 5$  enters the single-hop regime); we make no claims beyond that boundary.

5.3. **Methodological value.** This paper establishes a *precise boundary* for a natural approach to Weil positivity. The boundary is not a dead end:

- (1) It rules out the per-window hardcoded method as a route to full-spectrum RH verification, and identifies the mechanism (growing  $c_L$ , per-prime negative contributions, non-dominant cross terms).
- (2) The second-window cross-prime coupling  $J(\tau_2, \tau_3)$ , certified nonzero here for the first time, is a qualitatively new algebraic structure whose analysis may be useful independent of the positivity question.
- (3) The adjudication methodology—outsourcing conjectures to independent recomputation and reporting the refutations as results—is itself a contribution to the certification workflow (see `PROOF_CONSTITUTION.md`).

**Remark 5.2** (Honest negative result). A negative result that precisely characterises where a method fails and why is more useful to the community than a partial positive result that leaves the boundary ambiguous. Future work need not build window-3 machinery only to discover the same mechanism; the termination is already visible in window 2.

## APPENDIX A. DATA PROVENANCE TABLE

Every numerical statement in the body of the paper is traceable to one of the following entries.

### Integrity notes.

- P5 carries a `SIGN_FLIP` flag at  $d = 25$ : the  $d = 25$  value is  $-1.04 \times 10^{-9}$  and  $d = 30$  gives  $+8.85 \times 10^{-9}$ . The sign is not robustly positive until  $d = 30$ . This point has not been certify-graded. The paper text accurately reflects this.
- P7 documents the  $\lambda_{\text{lb}}$  vs.  $\lambda_{\text{min}}^{\text{true}}$  separation:  $\lambda_{\text{lb}}(0.42)_{\text{even}} = 9.31 \times 10^{-10}$  (certificate relaxation) while  $\lambda_{\text{min}}^{\text{true}}(0.42)_{\text{even}} = +3.19 \times 10^{-5}$  (**[pilot]**)—a ratio of  $\sim 34,000\times$ . The matrix is not negative-definite at  $L = 0.42$ ; the certificate merely became loose.
- “certify anchor” means: the float-center eigenvalue was reproduced by the trusted Arb-interval assembly to all printed digits; the pivot is separately certified by interval arithmetic (straddle confirmed). The eigenvalue itself is not an interval-enclosure result.



| THE RANGE OF A FINITE-SCALE WEIL POSITIVITY METHOD:SUCCESS IN THE FIRST PRIME WINDOW AND IT |                                                            |                        |      |                                                                                         |  |  |  |  |                       |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------|------|-----------------------------------------------------------------------------------------|--|--|--|--|-----------------------|
|                                                                                             | certify-indeterminate                                      |                        |      |                                                                                         |  |  |  |  |                       |
| P6                                                                                          | certify-indeterminate status, right end                    | $\approx 0.548$        | odd  | interval width $\sim 2.7 \times 10^{-4} \gg \text{signal} \sim 10^{-9}$                 |  |  |  |  | certify-indeterminate |
| P7                                                                                          | $\lambda_b$ vs. $\lambda_{\min}^{\text{true}}$ at $L=0.42$ | 0.42                   | even | $\lambda_b = 9.31 \times 10^{-10}, \lambda_{\min}^{\text{true}} = +3.19 \times 10^{-5}$ |  |  |  |  | [pilot]               |
| P8                                                                                          | $\max  \Delta C $ cross                                    | 3/5                    | even | $\geq 0.630$                                                                            |  |  |  |  | [certify]             |
| P9                                                                                          | $\max  \Delta C $ cross                                    | 3/5                    | odd  | $\geq 0.140$                                                                            |  |  |  |  | [certify]             |
| P10                                                                                         | min-pivot (full)                                           | 3/5                    | even | $[-0.01970, -0.01968]$                                                                  |  |  |  |  | [certify]             |
| P11                                                                                         | $\lambda_{\min}^{\text{true}}$ (trusted assembly)          | 3/5                    | even | -0.231                                                                                  |  |  |  |  | [pilot]               |
| P12                                                                                         | Layer-1 eig (refuted; reported +0.077, true -0.231)        | 3/5                    | even | refuted                                                                                 |  |  |  |  | [pilot] refuted       |
| P13                                                                                         | $\Delta V$ cross delay test                                | 0.552                  | even | -2.744                                                                                  |  |  |  |  | [pilot]               |
| P14                                                                                         | $\Delta V$ cross delay test                                | 0.580                  | even | -0.403                                                                                  |  |  |  |  | [pilot]               |
| P15                                                                                         | $\Delta V$ cross delay test                                | 0.600                  | even | -2.251                                                                                  |  |  |  |  | [pilot]               |
| P16                                                                                         | $\Delta_{\tau_3}$ (odd pivot contribution of $p=3$ )       | 3/5                    | odd  | -0.113                                                                                  |  |  |  |  | [pilot]               |
| P17                                                                                         | $\Delta_{\text{cross}}$ (even pivot contribution)          | 3/5                    | even | +0.195                                                                                  |  |  |  |  | [pilot]               |
| P18                                                                                         | min-pivot odd                                              | 11/20                  | odd  | $[-2.815 \times 10^{-2}, -1.720 \times 10^{-2}]$                                        |  |  |  |  | [certify]             |
| P19                                                                                         | min-pivot even (indeterminate)                             | 11/20                  | even | $[-3.87 \times 10^{-4}, +2.43 \times 10^{-4}]$                                          |  |  |  |  | [certify]             |
| P20                                                                                         | $\lambda_{\min}^{\text{true}}$ odd (certify anchor)        | 69/100                 | odd  | -0.36911                                                                                |  |  |  |  | [certify] anchor      |
| P21                                                                                         | pivot straddle odd                                         | 69/100                 | odd  | $\approx -0.099$                                                                        |  |  |  |  | [certify]             |
| P22                                                                                         | $\lambda_{\min}^{\text{true}}$ even (certify anchor)       | 67/100                 | even | -0.28596                                                                                |  |  |  |  | [certify] anchor      |
| P23                                                                                         | Odd pilot curve (9 pts, monotone)                          | window 2               | odd  | $-0.118 \rightarrow -0.369$                                                             |  |  |  |  | [pilot]               |
| P24                                                                                         | Even pilot curve (7 pts)                                   | window 2               | even | $-0.093 \rightarrow -0.286$                                                             |  |  |  |  | [pilot]               |
| P25                                                                                         | $d$ -refinement (even basin not $d$ -robust)               | 0.66-0.69              | even | artefact                                                                                |  |  |  |  | [pilot]               |
| P26                                                                                         | $\Lambda_2$ window-2 right-end infimum                     | $\rightarrow \log 2^-$ | odd  | $\approx -0.37$                                                                         |  |  |  |  | [certify] anchor      |